Project Report: Titanic Survival Prediction

1. Executive Summary

This comprehensive report details the end-to-end data science workflow applied to the Titanic dataset. The project involves historical data analysis, rigorous data cleaning, and the deployment of a machine learning classifier to predict passenger survival. Using features like socioeconomic status, age, and family relationships, we achieved a predictive accuracy of nearly 80%.

2. Project Objectives & Problem Statement

The Titanic disaster remains one of the most infamous maritime events in history. While there was an element of luck involved in surviving, it appears some groups of people were more likely to survive than others. This project seeks to:

- Analyze the demographic and socioeconomic factors influencing survival.
- Perform exploratory data analysis (EDA) to visualize key trends.
- Build a robust Logistic Regression model to classify survival outcomes.

3. Dataset Description

The dataset used for this analysis was **Day 3 Dataset.csv**, containing 891 records.
After cleaning, 712 records remained for high-fidelity modeling.

Feature	Description
Survived	Survival (0 = No, 1 = Yes)
Pclass	Ticket class (1 = 1st, 2 = 2nd, 3 = 3rd)
Sex	Gender of the passenger
Age	Age in years

SibSp	Number of siblings / spouses aboard
Parch	Number of parents / children aboard
Fare	Passenger fare
Embarked	Port of Embarkation (C = Cherbourg, Q = Queenstown, S = Southampton)

4. Data Preprocessing & Feature Engineering

4.1 Data Cleaning

Raw data is rarely ready for modeling. In this project:

- **Dimensionality Reduction:** Irrelevant or redundant columns such as *deck*, *embark_town*, and *who* were dropped to prevent overfitting.
- **Null Management:** Missing values in the *age* column were significant. Rows with missing data were excluded to maintain the integrity of the statistical analysis.

4.2 Feature Engineering: FamilySize

Individual counts of siblings and parents often tell only part of the story. By creating the **FamilySize** feature (SibSp + Parch), we gained a clearer view of how traveling in a unit impacted evacuation priority and survival chances.

5. Exploratory Data Analysis (EDA)

EDA is the backbone of understanding why a model makes certain predictions. Below are the visual insights gained from the dataset:

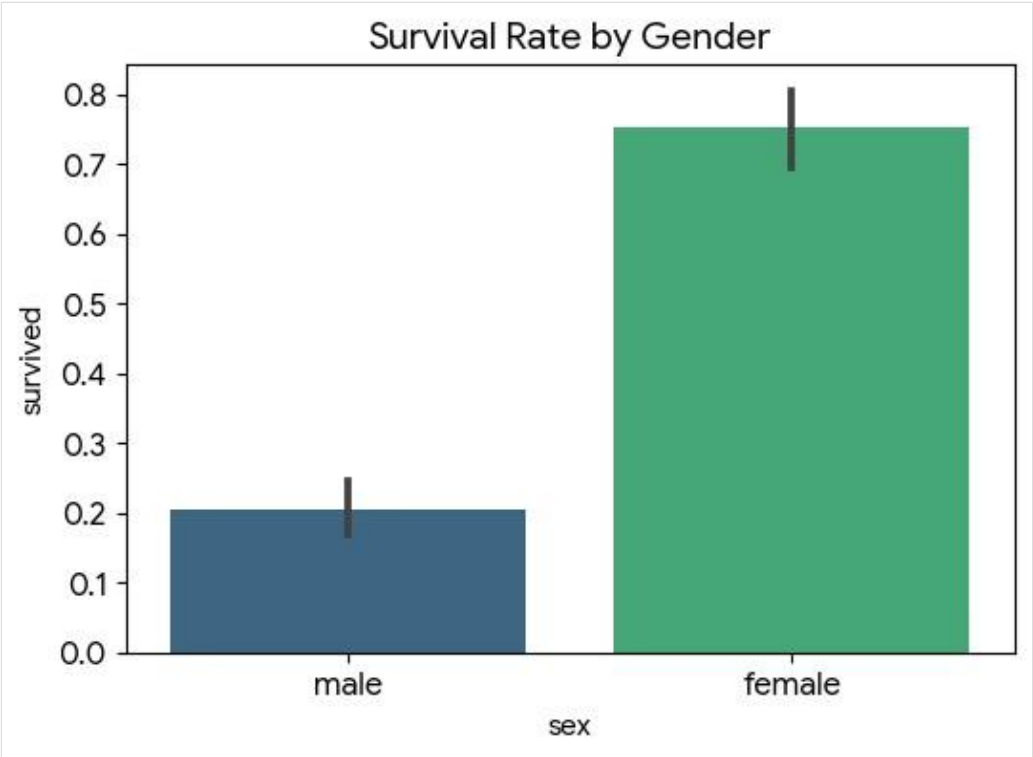


Figure 1: Survival rate strongly favored female passengers (over 75%) vs male passengers (~20%).

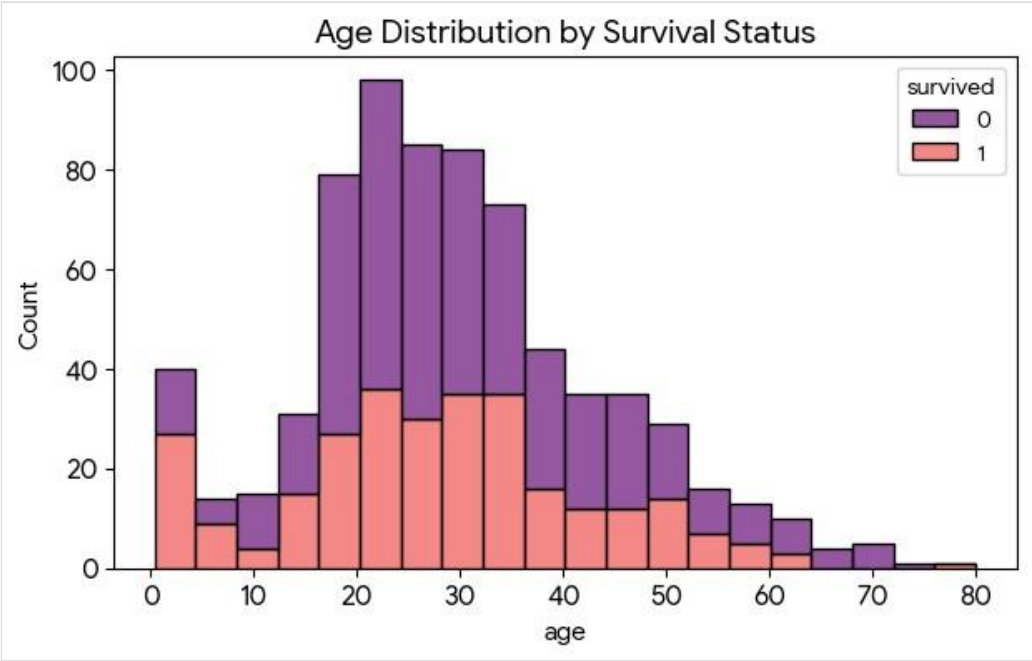
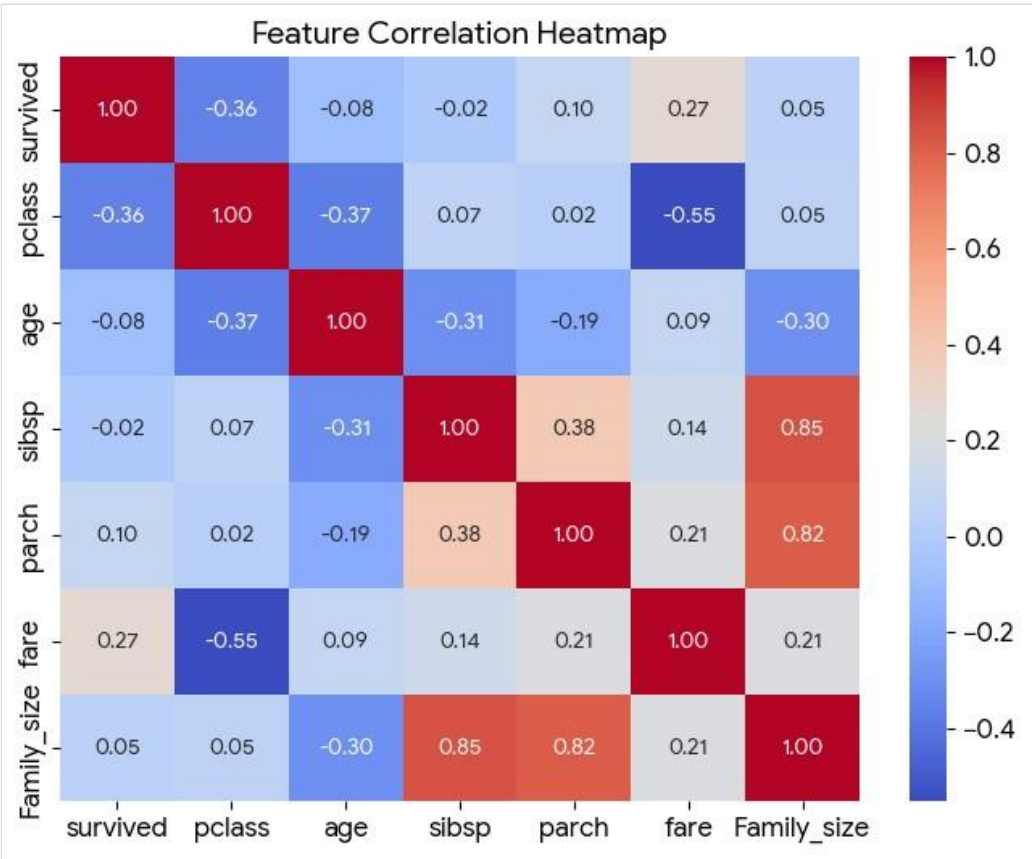


Figure 2: Stacked histogram showing that younger children had a distinct survival advantage.

6. Feature Correlation

The correlation heatmap highlights relationships between numeric variables. For instance, Pclass and Fare show a strong negative correlation, indicating that 1st class tickets (low class number) correspond to much higher fares.



7. Modeling & Technical Results

7.1 Logistic Regression Model

Logistic Regression was chosen for its interpretability and efficiency in binary classification. The model was trained using 80% of the cleaned data.

Model Performance Metrics:

- **Accuracy Score:** 79.72%
- **Precision (Survived):** 0.87
- **Recall (Survived):** 0.63
- **F1-Score:** 0.73

7.2 Confusion Matrix Analysis

The confusion matrix indicates the model's ability to distinguish between classes:

	Predicted: No	**Predicted: Yes**
Actual: No	74 (TN)	6 (FP)
Actual: Yes	23 (FN)	40 (TP)

The model successfully identified 40 survivors and 74 non-survivors in the test set.

8. Final Conclusion

This study demonstrates that survival on the Titanic was not random but heavily influenced by socioeconomic status (Pclass), gender (Sex), and age. The Logistic Regression model provides a reliable baseline for prediction, achieving nearly 80% accuracy. Future work could involve more complex algorithms like Random Forests or XGBoost to capture non-linear relationships further.